

Amlan Sarkar

Full-Stack Developer | AI-Integrated Systems | Frontend + Backend (Django)

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Professional Summary

Full-stack software developer with strong knowledge of object-oriented programming (OOP) and 3+ years of experience building production-ready applications. Skilled in designing, implementing, and debugging responsive user interfaces alongside scalable backend architectures using Python (Django), REST APIs, and relational databases like MySQL. Experienced in analyzing requirements to deliver complete product solutions, from AI-powered features (LLM, RAG) to real-time systems, ensuring high-quality, tested deliverables.

Technical Skills

Programming Languages: Python, Dart, C/C++, JavaScript

Core Concepts: Object-Oriented Programming (OOP), Data Structures & Algorithms, RESTful API Design, Clean Architecture, System Analysis

Backend & Databases: Django, MySQL, Firebase (Auth, Firestore, FCM), Server-side Logic

Frontend & Mobile: Flutter, Responsive UI, Material Design 3, MVVM

AI Systems & Tools: LLM Integration, Retrieval-Augmented Generation (RAG), Git, VS Code, Debugging & Testing Workflows

Education

BSc in Computer Science & Engineering

Expected July 2026

Khulna University of Engineering & Technology (KUET)

(Final semester; available immediately for remote work, and on-site/hybrid starting July 2026)

Selected Projects

AI-Powered Mental Health App

2024

Flutter, Firebase, Python (Django), LLM Integration

- Analyzing requirements and designing backend-integrated workflows for handling user context and AI response pipelines
- Designing and implementing a production-level application with AI-assisted interactions, ensuring correct functionality
- Integrating Firebase and backend services for authentication, real-time data handling, and persistent user workflows
- Debugging and testing modular frontend architecture to ensure scalability, maintainability, and smooth performance

Disaster Early Warning System

2025

Flutter, Firebase, REST APIs, MySQL

- Designed and implemented a real-time alert system with location-based warnings and push notifications
- Developed and tested backend-connected workflows for handling real-time updates and alert distribution
- Collaborated on an intuitive UI design for delivering critical information in time-sensitive scenarios

Urban Traffic Optimization App

2025

Flutter, REST APIs, Backend Integration

- Analyzed data requirements and developed a route-aware traffic application with real-time API integration
- Implemented structured communication between frontend UI and backend services
- Debugged and optimized performance for high-frequency data updates and user interactions

SafeSearch Engine for Kids

2025

Flutter, Python, Django, Machine Learning, REST APIs

- Designed and implemented an AI-powered search platform with content filtering and safe-result presentation
- Developed backend services using Python (Django-based architecture) for classification and API responses
- Tested RESTful APIs connecting frontend applications with backend intelligence systems to ensure reliable data flow

Achievements

- Built **10+ cross-platform applications** with real-time APIs and backend-connected workflows
- Delivered Android, iOS, Web, and Desktop applications from a shared architecture
- Built and integrated backend systems using Django and REST APIs in real-world projects